JUNIOR SCHOOL LEARNING HUB

GRADE 9 MATHEMATICS
END-OF-TERM 3 EXAMINATION

Name: _____	Adm. No.: _____	Date: _____	Class: _____
Time: 2 hours 30 minutes	Total: 100 marks	Term: 3	Learning Area: Mathematics

INSTRUCTIONS TO LEARNERS

- Answer all questions in Section A and Section B.
- Show all working clearly. Marks may be awarded for correct method.
- Use $\pi = 22/7$ where necessary unless otherwise stated.
- Use a non-programmable calculator only where permitted by your teacher.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. Simplify: $4a - 3(2a - 5)$. (2 marks)

2. Solve: $3(x - 2) = 2x + 7$. (2 marks)

3. Express 0.0000725 in standard form. (2 marks)

4. Factorise: $x^2 - 9x + 20$. (2 marks)

5. Find the value of y when $x = 4$ in $y = 3x^2 - 2x$. (2 marks)

6. Find the gradient of the line $y = 5x - 7$. (2 marks)

7. Find the sum of the interior angles of a hexagon. (2 marks)

8. A circle has radius 10.5 cm. Find its diameter. (2 marks)

9. Find the volume of a cuboid measuring 8 cm by 5 cm by 3 cm. (2 marks)

10. Convert 2.4 litres into millilitres. (2 marks)

11. Evaluate: $3^2 \times 3^3 \div 3^4$. (2 marks)

12. Find the median of 5, 8, 9, 12, 14, 17, 20. (2 marks)

13. A fair die is rolled once. Find the probability of obtaining an even number. (2 marks)

14. Make x the subject of the formula: $y = 4x + 9$. (2 marks)

15. Two angles on a straight line are $(3x + 10)°$ and $(2x - 5)°$. Find x. (2 marks)

16. A map scale is 1:25,000. Find the actual distance represented by 8 cm, in kilometres. (2 marks)

17. Find the simple interest on KSh 6,000 at 5% per annum for 3 years. (2 marks)

18. Find the next two terms: 2, 6, 12, 20, 30, … (2 marks)

19. A right-angled triangle has hypotenuse 13 cm and one shorter side 5 cm. Find the other side. (2 marks)

20. A learner answers 36 questions correctly out of 45. Find the percentage score. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Algebraic expressions and equations (10 marks)

a) Expand and simplify: $(x + 5)(x - 3)$. (3 marks)

b) Factorise completely: $3x^2 - 12x$. (2 marks)

c) Solve the equation: $x^2 + 2x - 15 = 0$. (3 marks)

d) If $f(x) = 2x - 7$, find $f(6)$. (2 marks)

22. Linear graphs and relationships (10 marks)

The equation of a straight line is $y = -x + 4$.

a) Complete a table of values for $x = -1, 0, 1, 2, 3$. (3 marks)

b) Draw the graph of $y = -x + 4$ for $-1 \leq x \leq 3$. (4 marks)

c) State the gradient and y-intercept. (2 marks)

d) Find the value of y when $x = 5$. (1 mark)

23. Geometry and mensuration (10 marks)

A closed cylindrical container has radius 7 cm and height 10 cm. Use $\pi = 22/7$.

a) Calculate the volume of the container. (3 marks)

b) Calculate the area of its curved surface. (3 marks)

c) Find the total surface area of the closed cylinder. (4 marks)

24. Statistics and probability (10 marks)

The number of books read by eight learners in a month was: 2, 3, 4, 3, 5, 6, 3, 6.

a) Construct a frequency table for the data. (3 marks)

b) Calculate the mean number of books. (3 marks)

c) State the mode and range. (2 marks)

d) A card is selected at random from cards numbered 1 to 10. Find the probability of selecting a prime number. (2 marks)

25. Similarity, scale drawing and Pythagoras' theorem (10 marks)

a) Two similar shapes have corresponding lengths in the ratio 3:4. If a side of the smaller shape is 12 cm, find the corresponding side of the larger shape. (3 marks)

b) A rectangular field is 24 m long and 10 m wide. Find the length of its diagonal. (4 marks)

c) A scale drawing uses 1 cm to represent 2.5 m. Find the drawing length for an actual distance of 15 m. (3 marks)

26. Financial mathematics and measurement (10 marks)

a) A trader buys a radio for KSh 3,200 and sells it for KSh 3,680. Calculate the percentage profit. (3 marks)

b) KSh 8,000 is invested at 6% simple interest per annum for 2.5 years. Calculate the interest and total amount. (3 marks)

c) A water tank contains 1.8 m^3 of water. Convert this volume into litres. (2 marks)

d) A square floor has a side of 6.5 m. Find its area. (2 marks)

MARKING SCHEME

Grade 9 Mathematics – End-of-Term 3 Examination

Total: 100 marks | Section A: 40 marks | Section B: 60 marks

SECTION A (40 MARKS)

1. $4a - 6a + 15 = 15 - 2a$. (2)

2. $3x - 6 = 2x + 7$; $x = 13$. (2)

3. 7.25×10^{-5}. (2)

4. $(x - 5)(x - 4)$. (2)

5. $y = 3(4^2) - 2(4) = 48 - 8 = 40$. (2)

6. 5. (2)

7. $(6 - 2) \times 180° = 720°$. (2)

8. $2 \times 10.5 = 21$ cm. (2)

9. $8 \times 5 \times 3 = 120$ cm^3. (2)

10. 2,400 ml. (2)

11. 3^(2+3−4) = 3. (2)

12. 12. (2)

13. 3/6 = 1/2. (2)

14. $x = (y - 9)/4$. (2)

15. $3x + 10 + 2x - 5 = 180$; $5x + 5 = 180$; $x = 35$. (2)

16. $8 \times 25{,}000 = 200{,}000$ cm = 2 km. (2)

17. I = 6,000 × 5 × 3/100 = KSh 900. (2)

18. 42, 56. (2)

19. Other side = $\sqrt{(13^2 - 5^2)} = \sqrt{144} = 12$ cm. (2)

20. 36/45 × 100 = 80%. (2)

SECTION B (60 MARKS)

21. Algebraic expressions and equations (10 marks)

$x^2 - 3x + 5x - 15 = x^2 + 2x - 15$. (3)

$3x(x - 4)$. (2)

$(x + 5)(x - 3) = 0$; $x = -5$ or $x = 3$. (3)

$f(6) = 12 - 7 = 5$. (2)

22. Linear graphs and relationships (10 marks)

y values: 5, 4, 3, 2, 1. (3)

Correct axes and scale (1); at least three correct points (2); accurate straight line (1). (4)

Gradient = −1; y-intercept = 4. (2)

$y = -5 + 4 = -1$. (1)

23. Geometry and mensuration (10 marks)

$V = \pi r^2 h = 22/7 \times 7 \times 7 \times 10 = 1{,}540\ cm^3$. (3)

$CSA = 2\pi rh = 2 \times 22/7 \times 7 \times 10 = 440\ cm^2$. (3)

$TSA = 2\pi rh + 2\pi r^2 = 440 + 308 = 748\ cm^2$. (4)

24. Statistics and probability (10 marks)

Values 2, 3, 4, 5, 6 with frequencies 1, 3, 1, 1, 2 respectively. (3)

Total = 32; mean = 32 ÷ 8 = 4. (3)

Mode = 3; range = 6 − 2 = 4. (2)

Primes are 2, 3, 5, 7; probability = 4/10 = 2/5. (2)

25. Similarity, scale drawing and Pythagoras' theorem (10 marks)

Scale factor = 4/3; larger side = 12 × 4/3 = 16 cm. (3)

$d^2 = 24^2 + 10^2 = 576 + 100 = 676$; $d = 26$ m. (4)

15 ÷ 2.5 = 6 cm. (3)

26. Financial mathematics and measurement (10 marks)

Profit = 480; percentage profit = 480/3,200 × 100 = 15%. (3)

I = PRT/100 = 8,000 × 6 × 2.5/100 = KSh 1,200; amount = KSh 9,200. (3)

$1\ m^3 = 1{,}000$ litres; $1.8\ m^3 = 1{,}800$ litres. (2)

Area = 6.5 × 6.5 = 42.25 m^2. (2)

ADMINISTRATION NOTE

This is an original teacher-prepared assessment resource for practice and school-based assessment. Before administration, the Mathematics teacher should verify that the assessed concepts match the actual Grade 9 Term 3 teaching sequence and the school scheme of work. Accept mathematically equivalent methods and correct units where appropriate.

Prepared by: Junior School Learning Hub